MICHAEL CHRISTWIN

Software Engineer | React, TypeScript, Python

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SUMMARY

Software engineer with hands-on experience building and deploying web applications across frontend and backend systems. Strongest in React and TypeScript, with experience in blockchain-enabled applications, data visualization, API integrations, and Python backend services. Currently, deepening backend engineering, database, networking, testing, and systems fundamentals.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, HTML, CSS, SQL
Frontend: React, React Router, SvelteKit, TanStack Router, TanStack Query, Tailwind CSS, shadcn/ui
Backend & Data: FastAPI, SQLAlchemy, SQLite, Redis, tRPC, REST, GraphQL, OpenAPI
Web3: wagmi, viem, smart contract integrations, Arweave
Infrastructure & Tooling: Docker, Git, Vercel, Cloudflare, Render, Turborepo, Bun

EXPERIENCE

Switch Electric / M3tering Protocol — *Software Developer, Project-based / Ongoing Collaboration*

- Built M3terscan, a protocol explorer that visualizes M3tering activity through tables, charts and heat-maps; developed a FastAPI aggregation service using SQLAlchemy, Alembic, SQLite, GraphQL integrations and Redis.
- Fetched protocol data streamed to Arweave, aggregated readings into hourly datasets, exposed frontend-oriented API responses, and cached responses in Redis for 24 hours; containerized the API with Docker and deployed it to Render.
- Built the M3terscan frontend with React, Tailwind CSS, shadcn/ui and TanStack Query, and adapted the same API into an Alliance Power dashboard for partner-specific meter and energy data.
- Developed M3trs interfaces for tokenizing, buying and selling meter yields using SvelteKit and React; integrated a generated OpenAPI client and a Bun + tRPC layer for OpenSea REST access and Arweave NFT image/metadata uploads.
- Built XCharge, a React application for purchasing energy credits on Gnosis Chain using xDAI, integrating wagmi, viem, contract ABIs and transaction-status handling.
- Built a Next.js World Chain miniapp using the project's custom SDK and established integration patterns, including frontend flows and API routes for authentication and transaction initiation.
- Developed and maintained public-facing protocol, token and company websites; integrated media assets, moved suitable resources to CDN delivery, configured static pre-rendering/Vercel deployment, and contributed to iterative SEO optimization.

Learnhall — *Frontend Developer (Volunteer) | Dec 2022 – Aug 2023*

- Implemented a redesign of Learnhall's public-facing website using React and Tailwind CSS, collaborating with designers and other developers to translate supplied designs into responsive interfaces.

Switch Electric — *Frontend Development Intern | Oct 2022 – Jan 2023*

- Built small frontend projects with React and Bootstrap and re-implemented an existing HTML/CSS/JavaScript NFT marketplace interface using React and component-based UI patterns.

SELECTED PROJECTS

M3terscan — React, FastAPI, SQLAlchemy, Redis, Docker, GraphQL
Protocol-data aggregation and visualization platform with a Python API, caching layer and interactive frontend.
M3trs — React, SvelteKit, tRPC, OpenAPI, TanStack Query, Arweave, Docker
Web3 application for tokenizing and trading meter yields with type-safe frontend/API integration and decentralized metadata storage.
XCharge — React, TypeScript, wagmi, viem, Gnosis Chain
On-chain energy-credit purchasing interface with smart-contract transaction execution and status tracking.

EDUCATION

Federal Government College Enugu | Sep 2016 – Oct 2022